



Medical Microbiology

Lecture Seven

Normal Microbial Flora

Third stage- College of Dentistry

Dr. Dheyaa Naji Hamzah

Important terms:

1. **Mutualism (Synergism)** : Both members of the association are benefit.
2. **Commensalism**: There is no apparent benefit or harm to either member of the association.
3. **Parasitism (antagonism)**: One member of the association lives at the expense of the other member.
4. **Opportunity** : A commensal organism that can cause disease in certain circumstances.

What are the Normal Flora?

- The human body is home to trillions of microorganisms, including bacteria, fungi, and viruses.
- These microorganisms are collectively called the **normal flora** or **microbiota**.
- Most of them live in a **symbiotic** or **commensal** relationship with the host.
- Normal flora play an essential role in maintaining health and preventing disease.
- Internal tissues, e.g. **blood, brain, muscle**, are normally **free** of microorganisms.



Microbiota Development in Infants:

- At birth, the oral cavity is completely sterile. Soon after delivery, **Streptococcus salivarius** becomes the most dominant organism, forming nearly **98%** of the early oral microbiota.
- **Neonatal period:**
- During the first weeks of life, the mouth is mainly inhabited by **aerobic bacteria** and **Candida albicans**.
- **First year of life:**
- As the infant's teeth begin to erupt, new ecological surfaces appear, allowing colonization by organisms such as **Streptococcus mutans** and **Streptococcus sanguis**.
- These bacteria require **hard, non-epithelial surfaces** like, teeth—in order to attach and grow.

Normal Oral Flora :

- The presence of **nutrients**, **epithelial debris**, and **secretions** make the mouth a favorable habitat for a great variety of bacteria.
- Oral bacteria include ***Streptococci***, ***Lactobacilli***, ***Staphylococci*** and, ***Actinomyces*** with a great number of anaerobes, especially ***Bacteroides***.
- The composition varies with age, diet, and oral hygiene.

Types of Normal Flora

Normal flora can be classified into:

1-Resident Flora:

- Permanently colonize specific body sites.
- Usually harmless and beneficial.
- Example: *Streptococcus salivarius* in the mouth.

2-Transient Flora:

- Temporarily present; come and go.
- May include potential pathogens.
- Example: *Staphylococcus aureus* on the skin surface.

Functions of Normal Flora:

1-Protective Role:

- Compete with pathogens for nutrients and adhesion sites.
- Produce bacteriocins that inhibit harmful microbes.

2-Metabolic Role:

- Synthesize vitamins (e.g., Vit. K, Vit. B12).
- Help in digestion of complex carbohydrates.

3-Immunological Role:

- Stimulate the immune system to develop properly.
- The normal flora behave as antigens, they induce an immunological response.

4-Normal flora stimulate the development of certain tissues.

- Ex. certain lymphatic tissues (Peyer's patches) in the GI tract.

Maintaining a Healthy Microbiota:

- 1-Good oral hygiene.
- 2-Rational antibiotic use.
- 3-Balanced diet rich in fiber and low in sugar.
- 4-Regular dental checkups.

Why Microbial Flora colonize in particular tissue?

1-Tissue Tropism:

One possible explanation for tissue tropism is that the host supplies specific growth factors required by the bacterium to survive and multiply.

2-Extracellular Compartments:

Some bacteria specifically colonize certain tissues by using surface structures such as capsules, fimbriae, or cell wall components as ligands that bind to receptors on host cells.

3-Biofilm Formation:

Certain bacteria can form **biofilms** on tissue surfaces, allowing them to adhere, survive, and establish stable microbial communities.

The Role of Saliva in Maintaining a Healthy Oral Environment :

- Saliva helps preserve oral health through several key functions:

1-PH Regulation:

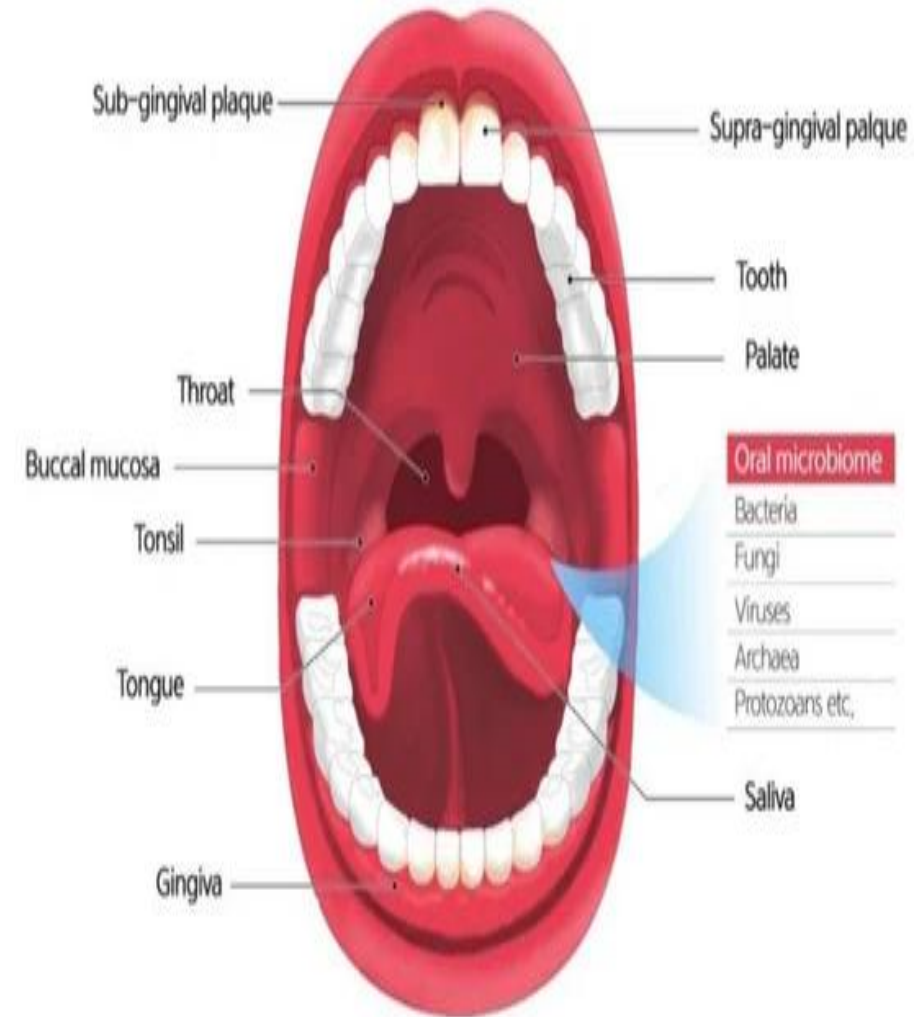
Saliva acts as a natural buffer, stabilizing the oral pH and creating favorable conditions for the normal microbial community.

2-Nutrient Source:

Many oral microorganisms rely on salivary proteins and glycoproteins as an important source of nutrients for their growth.

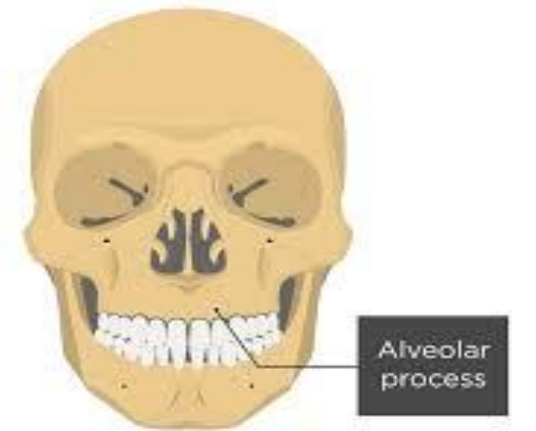
3-Pellicle Formation:

Salivary proteins and glycoproteins also contribute to the creation of the **acquired enamel pellicle**, a protective layer that forms on tooth surfaces.



Oral Flora as a Source of Oral and Systemic Disease:

- The normal oral microbiota is often responsible for a variety of oral conditions, including **dental caries, gingivitis, periodontal disease, and oral abscesses**.
- When these microorganisms gain access to deeper tissues, they may cause abscesses in sites such as the **alveolar bone, or even the brain**.
- Additionally, if **oral streptococci** enter the bloodstream particularly through tissue injury during dental treatments, they can adhere to **heart valves** and potentially initiate **sub acute bacterial endocarditis**.



Factors Influencing Microbial Growth in the Oral Cavity:

Several factors determine how microorganisms grow and survive in the oral environment, including:

1-Temperature: Optimal warmth supports microbial activity.

2-Redox potential / Oxygen availability: Levels of anaerobiosis influence which species can thrive.

3-pH: Acidity or alkalinity affects microbial balance and growth.

4-Nutrient supply (diet): Dietary components provide essential nutrients for microbes.

5-Host defenses: Both innate and adaptive immune mechanisms regulate microbial populations.

6-Antimicrobial substances and inhibitors: Medications and natural inhibitors in the mouth can restrict microbial growth.

Distribution of Normal Flora in the Body

Body Site	Common Microorganisms	Notes
Skin	Staphylococcus epidermidis	Prevent colonization by pathogens.
Nasal cavity	Staphylococcus aureus, Streptococcus spp.	Some are opportunistic pathogens.
Oral cavity	Streptococcus mutans, Actinomyces, Lactobacillus	Important in dental plaque and caries.
Intestine	E. coli, Bacteroides, Lactobacillus	Aid digestion and vitamin synthesis.
Vagina	Lactobacillus spp.	Maintain acidic pH to prevent infection.



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Any Question?



Thank you